

Medical Readmission Risk Assessment Report

Humber College Institute of Technology & Advanced Learning
Capstone Project - Computer Programming
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Patient Information

Field	Value
Age Group	[50-60)
Gender	Female
Time in Hospital (days)	5
Admission Type	1
Discharge Disposition	1
Admission Source	1
Number of Medications	10
Lab Procedures	30
Medical Procedures	2
Number of Diagnoses	3
Previous Inpatient Visits	1
Previous Outpatient Visits	2
Previous Emergency Visits	0
Diabetes Medication	Yes

Medication Change	Ch
A1C Test Result	Norm
Max Glucose Serum	None
Insulin	No
Metformin	Steady
Primary Diagnosis	250.00

Risk Assessment Results

Readmission Risk Score: 0.796
Risk Level: **High Risk**
Model Confidence: 79.6%

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AI-Generated Medical Insights

Based on the provided patient information, the primary concern appears to be the management of uncontrolled diabetes mellitus (diagnosis 250.00), which led to an emergency admission despite a normal long-term A1C result. This suggests an acute glycemic event rather than chronic poor control. The patient was discharged home, indicating stability at the time of discharge.

****Key Medical Insights & Recommendations:****

- **Diabetes Management:**** While the A1C is favorable, the "uncontrolled" primary diagnosis and emergency admission highlight the importance of strict adherence to the updated medication regimen (including steady metformin and any other changes made during hospitalization). Regular home blood glucose monitoring is crucial to prevent future acute episodes.
- **High Readmission Risk Mitigation:**** With an exceptionally high readmission risk score of 0.80, prompt and comprehensive post-discharge follow-up is paramount. This should include a scheduled visit with a primary care physician or endocrinologist soon after discharge to review current medication efficacy, reinforce diabetes self-management education (diet, exercise, sick day rules), and address any potential barriers to care to prevent readmission.

***Disclaimer:** This information is for general informational purposes only and is not intended to be a substitute for professional medical advice, diagnosis, or treatment. Always seek the advice of your physician or other qualified health provider with any

questions you may have regarding a medical condition.*

Understanding Your Risk Score

The readmission risk score is calculated using advanced machine learning algorithms that analyze multiple patient factors including medical history, current medications, diagnosis codes, hospital stay characteristics, and previous healthcare utilization patterns.

Risk Categories:

- **Low Risk (0.0-0.3):** Minimal likelihood of readmission
- **Medium Risk (0.3-0.7):** Moderate readmission risk requiring attention
- **High Risk (0.7-1.0):** Elevated readmission risk requiring intensive follow-up

IMPORTANT MEDICAL DISCLAIMER:

This report is generated by an AI system for informational purposes only and should not be considered medical advice, diagnosis, or treatment recommendations. Always consult with qualified healthcare professionals for medical decisions. The predictions are based on statistical models and may not account for all individual patient factors.

Generated by SweatHogs Medical AI Prediction System
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